The Simple Mediation Model

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2025-05-01

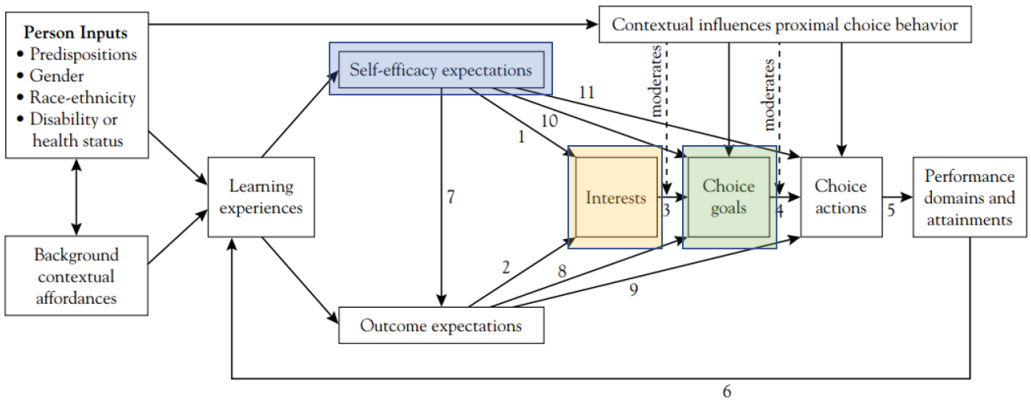
Problem

Why is it that some students have stronger intentions of enrolling in computer science courses whereas others don't?

Theory

According to social cognitive career theory, an individual's choice goal is predicted by both self-efficacy (path 10) and interest (path 3). Furthermore, self-efficacy predicts interest (path 1).

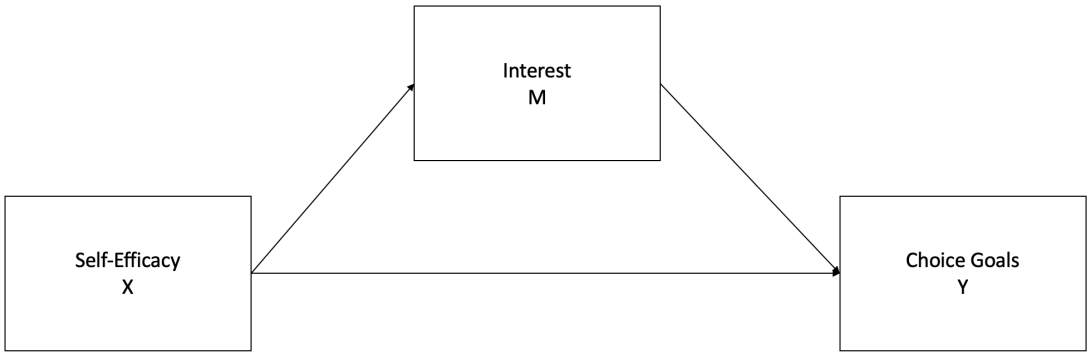
FIGURE 7.2. Model of Person, Contextual, and Experiential Factors Affecting Career-Related Choice Behavior



Note: Direct relations between variables are indicated with solid lines; moderator effects (where a given variable strengthens or weakens the relations between two other variables) are shown with dashed lines.
Source: Copyright © 1993 by R. W. Lent, S. D. Brown, and G. Hackett. Reprinted by permission.

We want to test if this part of this model is true or not. So, we go out to collect some data but what should we do next?

Notation: Diagram

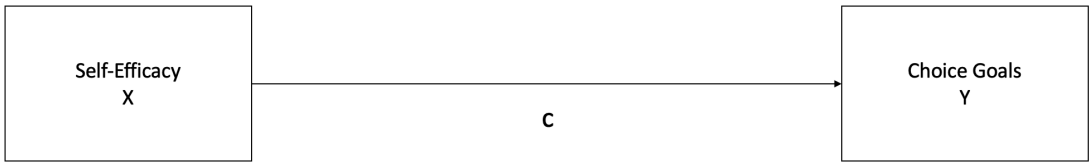


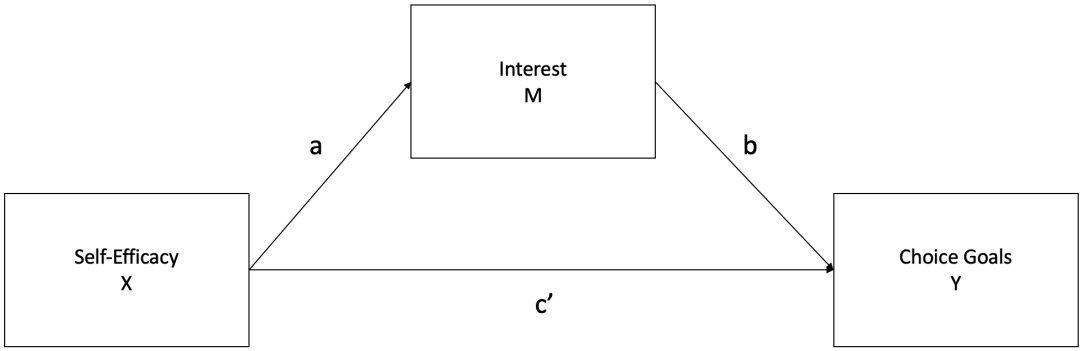
- **X**: causal antecedent variable
- **Y**: consequent variable
- **M** is sometimes called a "mediator", but it goes by other names as well.
- For M to be an intermediary, it must be located causally between X and Y. That is, M comes after X and comes before Y.
- A mediation model links a putative cause (X) to a presumed effect (Y) at least in part via an intermediary variable (M). An intermediary variable can be a psychological **state**, a cognitive **process**, an affective response, or any other conceivable **"mechanism"** through which X exerts an effect on Y. X affects M, which, in turn, affects Y.
- Mediator models are causal models and carry with them the usual criteria for making causal claims. These are often difficult to establish unequivocally, statistically or otherwise. We do our best, recognizing that sometimes our data do not lend themselves to causal claims. That is, theory is sometimes the sole foundation upon which our causal claims rest.

Activity

- Use the notation shown here to draw the relation you would like to investigate.

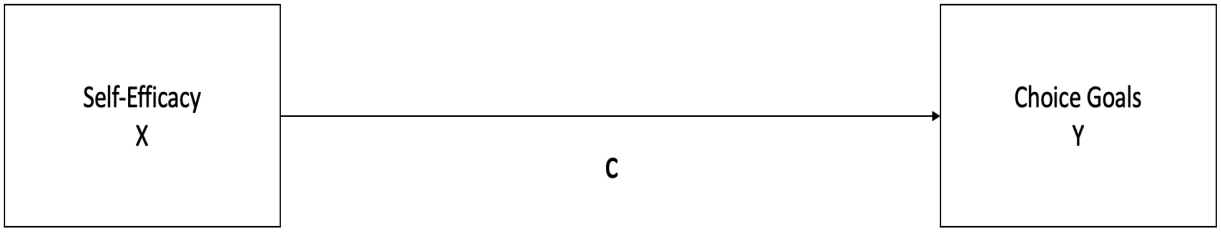
Notation: Total, direct, and indirect effects



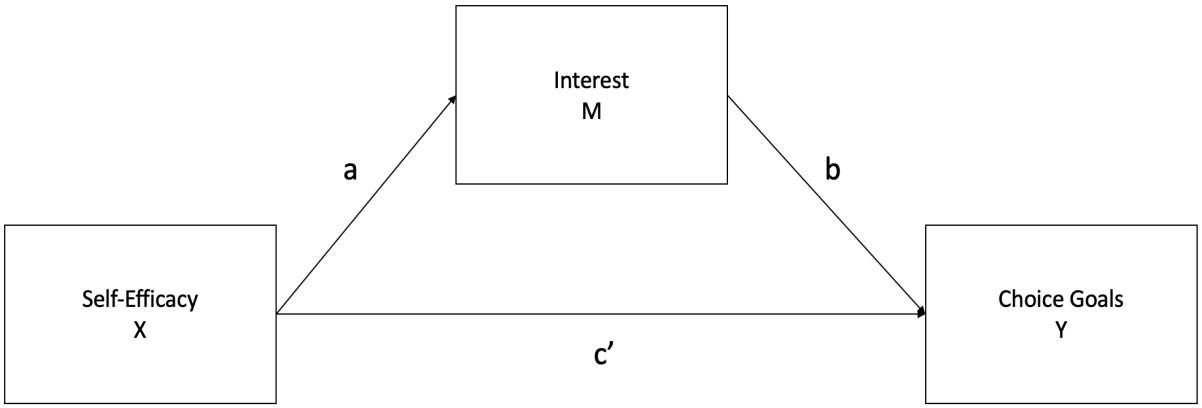


- Let a, b, c, and c' be quantifications of causal effects, such as regression coefficients in an ordinary least squares model (or maximum likelihood path estimates in a structural equation model).
- c = "total effect" of X on Y (i.e., the effect of X on Y when we don't account for the effect of M)
- **a × b = "indirect effect" of X on Y (This is our main focus.)**
- c' = "direct effect" of X on Y
- total effect (c) = direct effect (c') + indirect effect (a × b)
- indirect effect (a × b) = total effect (c) - direct effect (c')
- We can still do our mediation analysis even if the total effect is zero, **c is not the precursor of the mediation analysis**.

Regression Equation



$$ChoiceGoals_j = i + cSelf - Efficacy_j + \varepsilon_j$$



$$Interest_j = i + aSelf - Efficacy_j + \varepsilon_j$$

$$ChoiceGoals_j = i + bInterest_j + c'Self - Efficacy_j + \varepsilon_j$$

Run Process Macro

Whenever you would like to run conditional process analysis, don't forget to open and run the script of Process Macro!

```
******************** PROCESS for R Version 4.3.1 ********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

PROCESS is now ready for use.
Copyright 2020-2023 by Andrew F. Hayes ALL RIGHTS RESERVED
Workshop schedule at http://haskayne.ucalgary.ca/CCRAM
```

Load Packages and Import Data

```
library(tidyverse)
library(psych)
library(haven)

data <- read_sav("scct.sav")

#view our data
data
```

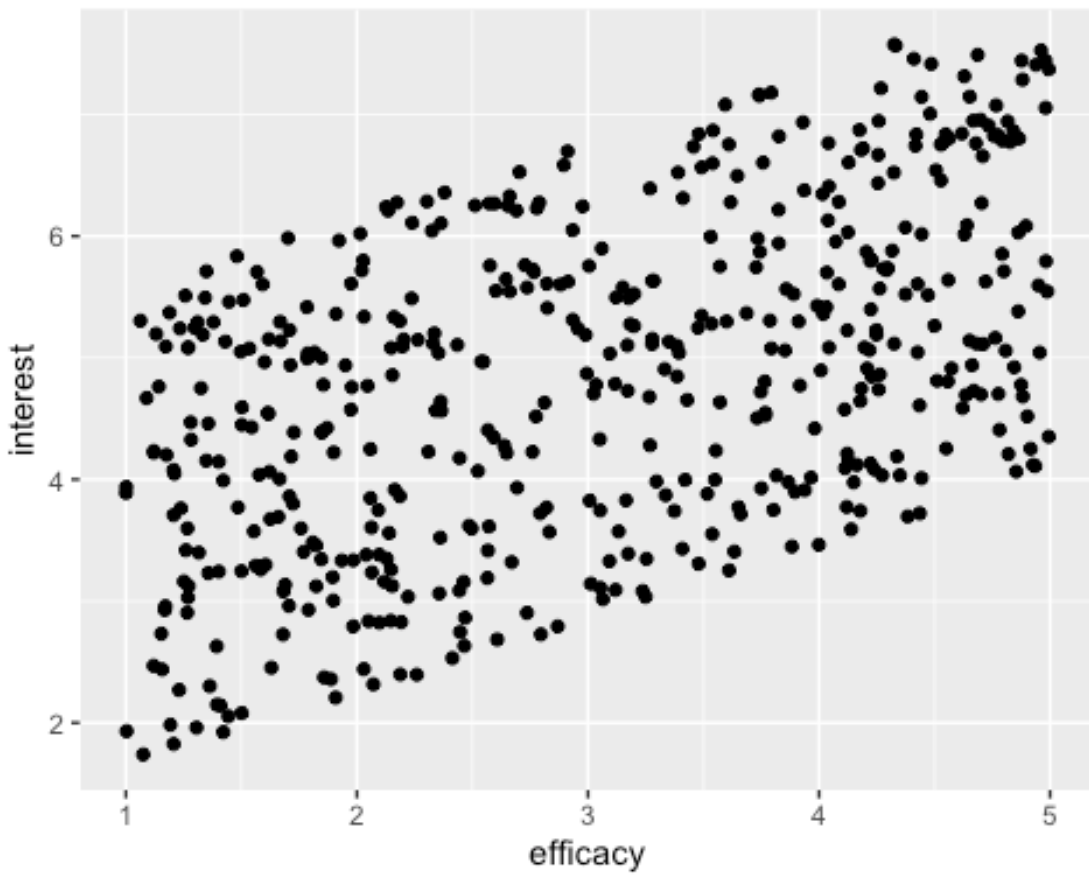
```
# A tibble: 500 × 3
   efficacy interest intention
      <dbl>    <dbl>     <dbl>
 1     3.13     3.57      4.94
 2     2.57     6.27      7.94
 3     2.32     6.04      4.16
 4     3.83     6.82      6.61
 5     4.25     5.23      8.12
 6     3.42     4.00      4.79
 7     4.68     6.76      6.72
 8     4.95     5.60      6.94
 9     1.92     5.96      7.84
10     4.44     3.72      4.07
# ℹ 490 more rows
```

```
#descriptive statistics
describe(data)
```

```
          vars   n mean   sd median trimmed  mad  min  max range  skew
kurtosis
efficacy     1 500 3.03 1.20   3.04    3.03 1.63 1.00 4.99  3.99 -0.01    -
1.34
interest     2 500 4.82 1.32   4.91    4.82 1.42 1.74 7.58  5.84 -0.05    -
0.71
intention    3 500 5.82 1.40   5.89    5.83 1.59 2.33 9.17  6.84 -0.04    -
0.67
            se
efficacy  0.05
interest  0.06
intention 0.06
```

Visualize

```
data |>
  ggplot(mapping = aes(x = efficacy, y = interest)) +
  geom_point()
```




```
data |>
  ggplot(mapping = aes(x = efficacy, y = intention)) +
  geom_point()
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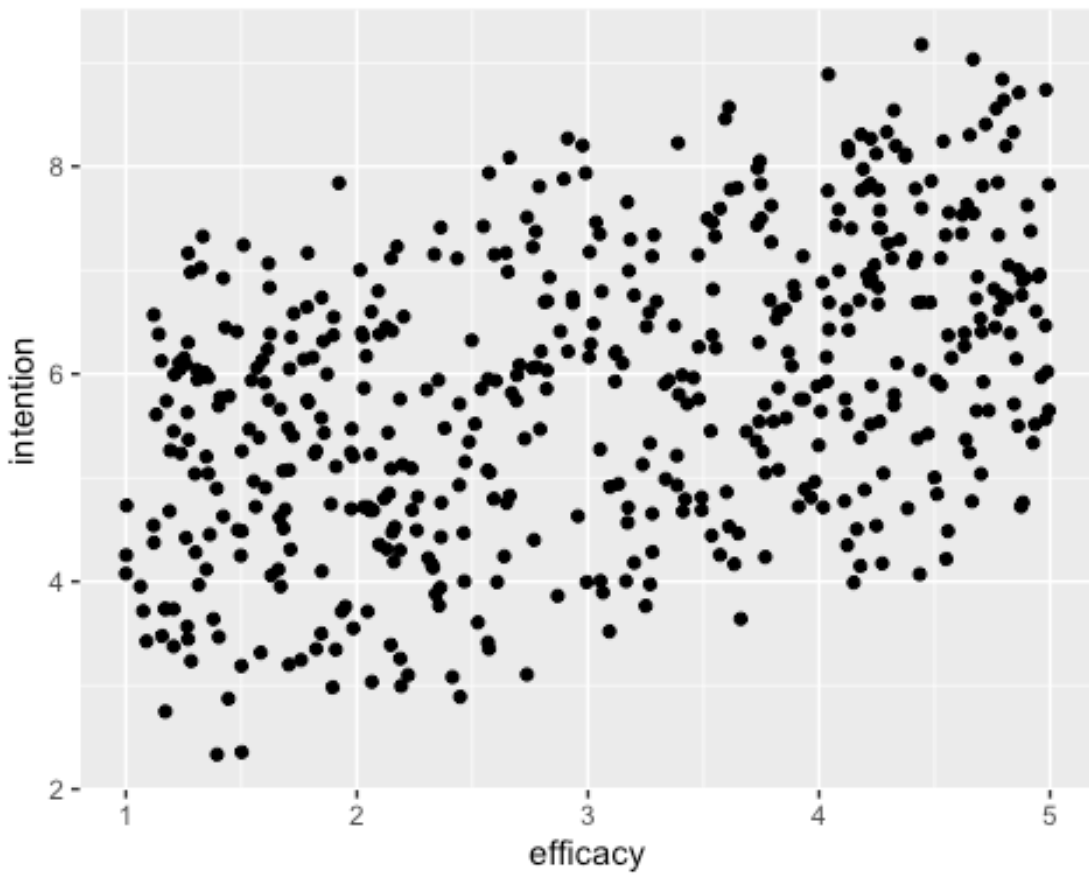


```
data |>
  ggplot(mapping = aes(x = interest, y = intention)) +
  geom_point()
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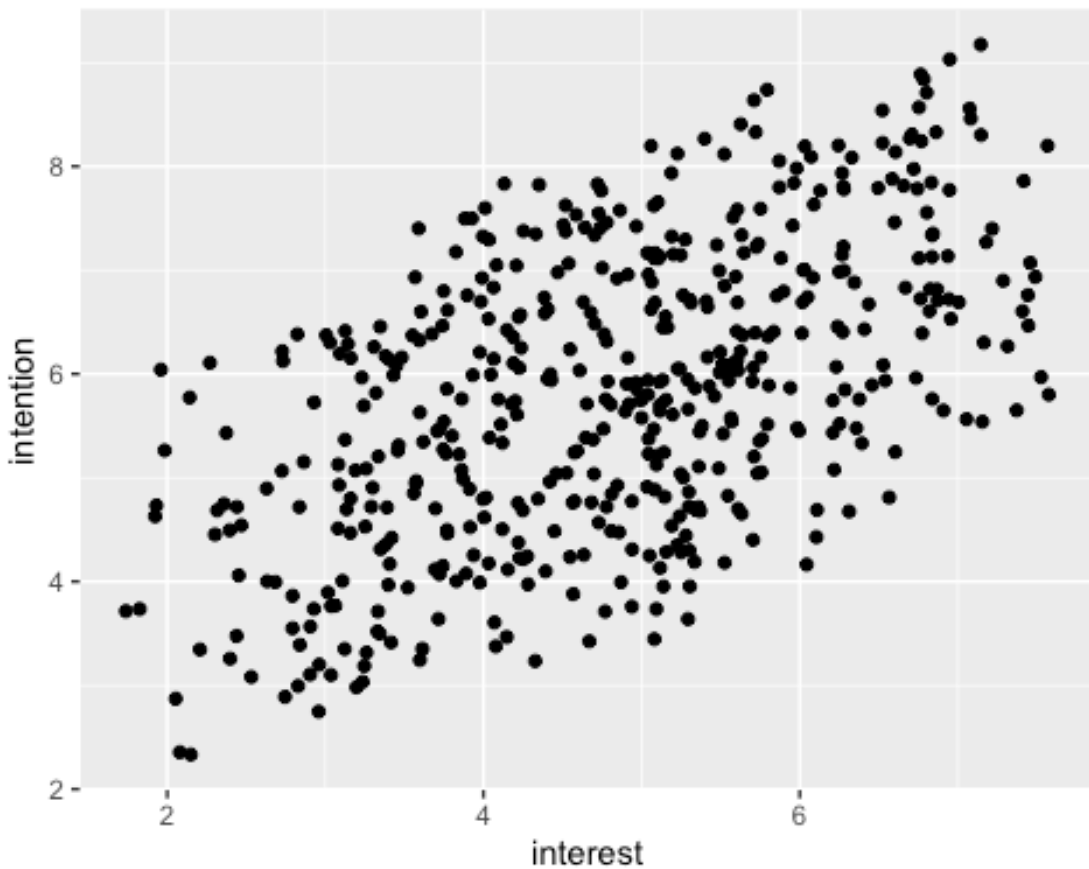

Model (condtional process analysis: the simple mediation model)

```
process(data=data,y="intention",x="efficacy",m="interest", total=1,model=4,
seed=12345)


******************** PROCESS for R Version 4.3.1 ********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

Model : 4
    Y : intention
    X : efficacy
    M : interest

Sample size: 500

Custom seed: 12345
```

```
**********************************************************************
Outcome Variable: interest

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
      0.5100     0.2601     1.2984   175.0417     1.0000   498.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     3.1164     0.1385    22.5079     0.0000     2.8444     3.3885
efficacy     0.5626     0.0425    13.2303     0.0000     0.4791     0.6462

**********************************************************************
Outcome Variable: intention

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
      0.5900     0.3480     1.2864   132.6592     2.0000   497.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     2.7686     0.1957    14.1436     0.0000     2.3840     3.1532
efficacy     0.2658     0.0492     5.4023     0.0000     0.1692     0.3625
interest     0.4667     0.0446    10.4631     0.0000     0.3791     0.5543

************************ TOTAL EFFECT MODEL **************************
Outcome Variable: intention

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
      0.4521     0.2044     1.5666   127.9669     1.0000   498.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     4.2230     0.1521    27.7662     0.0000     3.9242     4.5219
efficacy     0.5284     0.0467    11.3122     0.0000     0.4366     0.6202

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```
#overall model: Multiple R-squared (ranges from 0-1) helps us evaluate the
predictability of our model. The higher the better.

#Outcome Variable: interest => path a

#interpretation: About 28.15% of variance in interest is explained by self-
efficacy.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Self-efficacy significantly and positively predicts
interest. One unit increase in self-efficacy is expected to be associated
with an increase of 0.6395 units in interest.

#Outcome Variable: intention => path b and c'

#interpretation: About 31.28% of variance in intention to enroll in computer
science courses is explained by self-efficacy AND interest.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation b: Interest significantly and positively predicts intention.
One unit increase in interest is expected to be associated with an increase
of 0.3765 units in intention.

#interpretation c': Self-efficacy significantly and positively predicts
intention. One unit increase in self-efficacy is expected to be associated
with an increase of 0.3100 units in intention.

#Outcome Variable: intention => path c

#interpretation: About 20.44% of variance in intention to enroll in computer
science courses is explained by self-efficacy ALONE.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation c: Self-efficacy significantly and positively predicts
intention. One unit increase in interest is expected to be associated with an
increase of 0.5284 units in intention.

#TOTAL, DIRECT, AND INDIRECT EFFECTS OF X ON Y
#Indirect effect(s) of X on Y => a x b
#To evaluate whether M significantly mediates the relation between X and Y,
```

```
you need to check BootLLCI and BootULCI. If zero is not in the confidence
interval, then we can claim an indirect effect different from zero with 95%
confidence. This is akin to (though not exactly the same as) rejecting the
null hypothesis of no indirect effect at the alpha = 0.05 level of
significance.

#interpretation a x b: Interest significantly mediates the relation between
self-efficacy and intention (ab = 0.2626, 95% CI [.2041, .3273]).

#interpretation a x b: Self-efficacy is significantly and indirectly related
to intention through the pathway of interest (ab = 0.2626, 95% CI [.2041,
.3273]).

#Conclusion: The higher self-efficacy in studying computer science you have,
the more interested you are in studying computer science, which is, in turn,
signifincatly associated with a higher intention to enroll in computer
science courses.
```

Activity

Let's examine the relation between income, happiness, and health. Your theory is that income serves as a causal antecedent variable, happiness serves as a mediator, and health serves as a consequent variable.

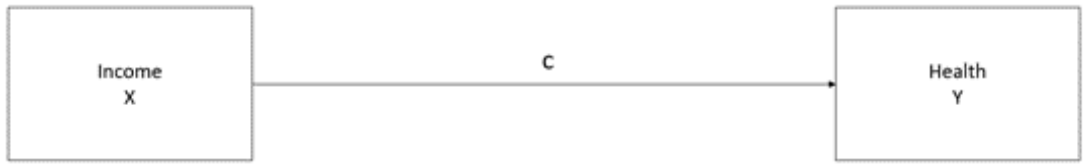
- Draw the diagram to represent the simple mediation model.
- Import the "health" dataset.
- Take a look at your data and descriptive statistics.
- Visualize the relation between income and happiness. Is it positive or negative?
- Visualize the relation between happiness and health. Is it positive or negative?
- Visualize the relation between income and health. Is it positive or negative?
- Run the conditional process analysis (model 4)
- Find out the effect of path a, b, c, c', and a x b.
- Evaluate the overall model and individual predictor.
- Write down your interpretation of the results for each path (i.e., a, b, c, c', and a x b)
- Is your theory correct? Does being happy significantly mediate the relation between income and health?

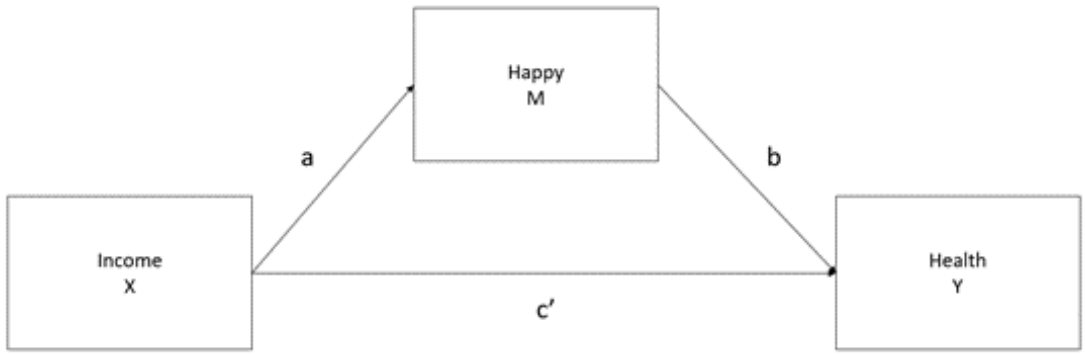
Solution

Run Process Macro

Please open and run your Process Macro script.

Diagram





Load Packages and Import Data

```
library(tidyverse)
library(psych)
library(haven)

data <- read_sav("health.sav")

#view our data
data

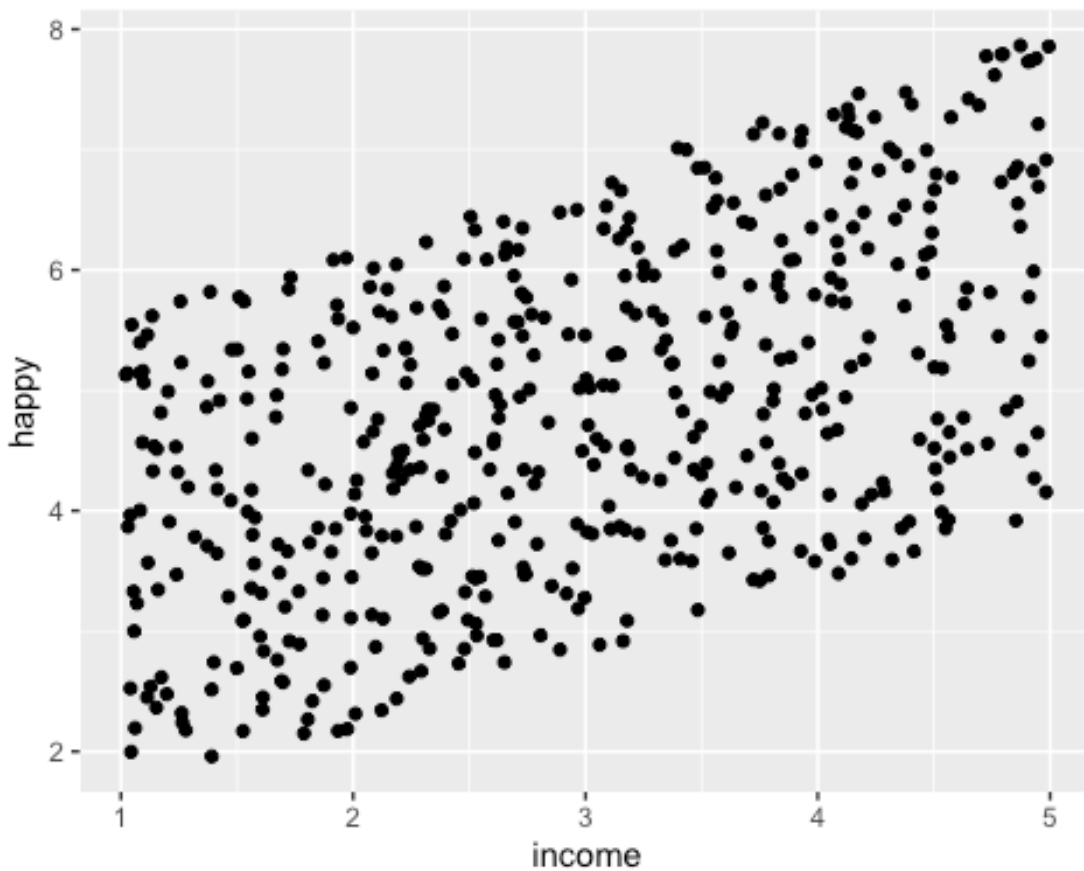
# A tibble: 500 × 3
   income happy health
    <dbl> <dbl>  <dbl>
 1   1.53  5.74   6.72
 2   2.55  5.59   4.82
 3   1.06  3.00   4.43
 4   3.48  3.85   3.36
 5   1.02  5.13   4.96
 6   3.51  6.85   4.92
 7   1.87  3.13   3.86
 8   4.48  6.15   5.79
 9   4.02  4.84   6.05
10   4.49  6.31   5.56
# ℹ 490 more rows

#descriptive statistics
describe(data)
```

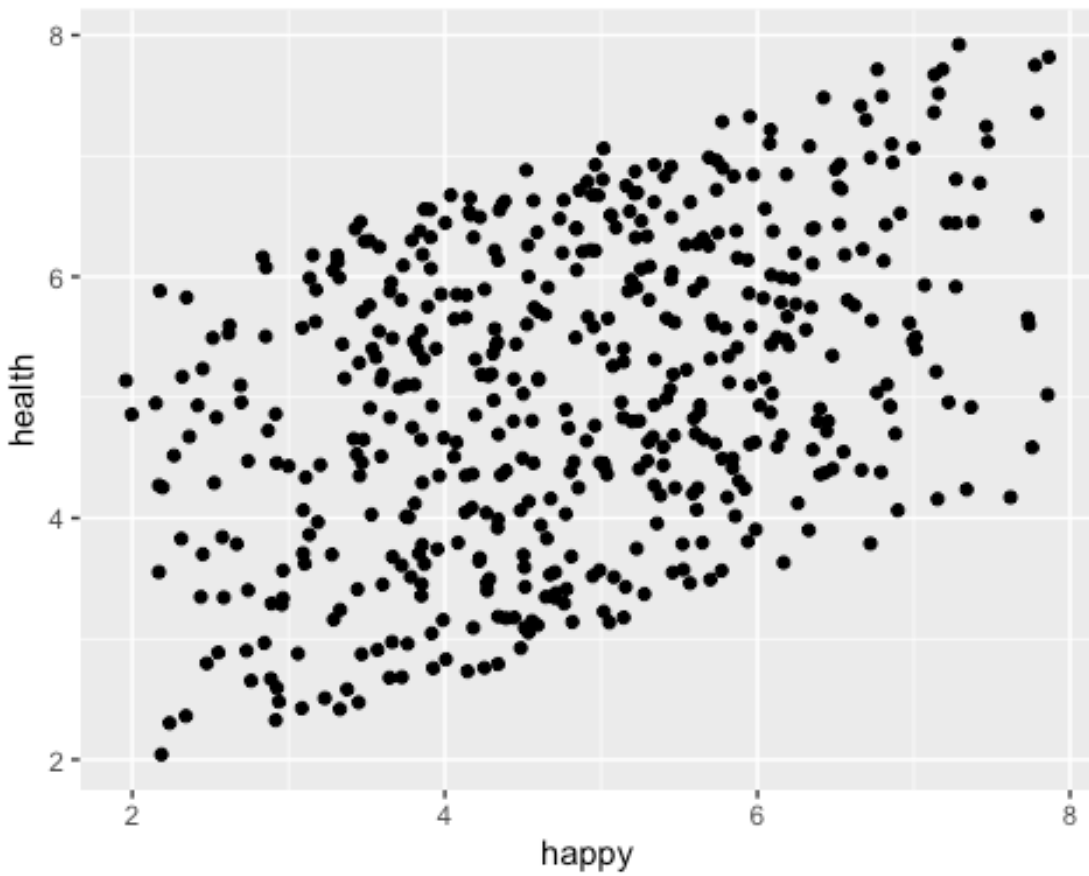
```
       vars   n mean   sd median trimmed  mad  min  max range  skew kurtosis
income    1 500 2.96 1.13   2.96    2.96 1.37 1.02 4.99  3.97  0.03    -1.13
happy     2 500 4.82 1.36   4.78    4.81 1.48 1.96 7.86  5.90  0.09    -0.72
health    3 500 5.05 1.28   5.10    5.07 1.50 2.04 7.92  5.88 -0.10    -0.83
         se
income 0.05
happy  0.06
health 0.06
```

Visualize

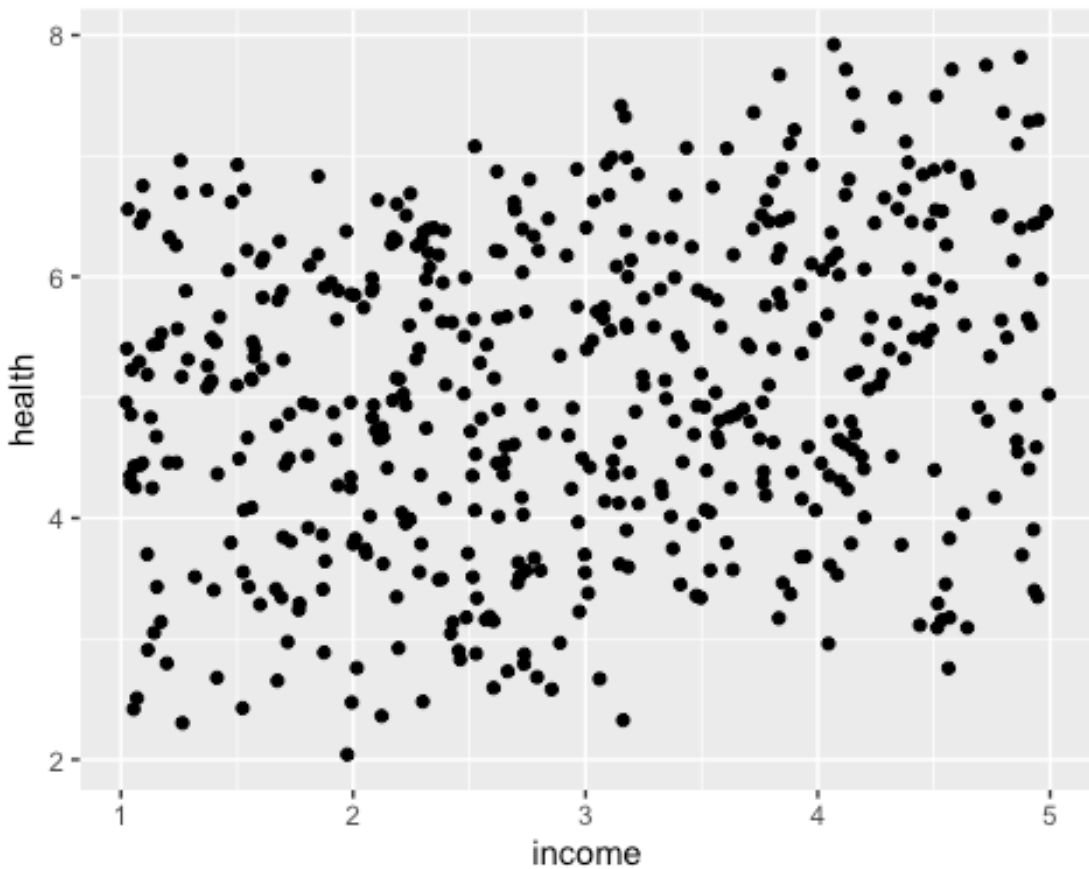
```
data |>
  ggplot(mapping = aes(x = income, y = happy)) +
  geom_point()
```




```
data |>
  ggplot(mapping = aes(x = happy, y = health)) +
  geom_point()
```




```r
data |>
  ggplot(mapping = aes(x = income, y = health)) +
  geom_point()
```



Model (condtional process analysis: the simple mediation model)

```
process(data=data,y="health",x="income",m="happy", total=1,model=4,
seed=12345)


******************** PROCESS for R Version 4.3.1 ********************

           Written by Andrew F. Hayes, Ph.D.  www.afhayes.com
   Documentation available in Hayes (2022). www.guilford.com/p/hayes3

**************************************************************************

Model : 4
    Y : health
    X : income
    M : happy

Sample size: 500

Custom seed: 12345
```

```
**********************************************************************
Outcome Variable: happy

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.5171     0.2674     1.3489   181.7738     1.0000   498.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     2.9775     0.1463    20.3519     0.0000     2.6901     3.2650
income       0.6221     0.0461    13.4824     0.0000     0.5315     0.7128

**********************************************************************
Outcome Variable: health

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.4166     0.1735     1.3566    52.1779     2.0000   497.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     3.1579     0.1986    15.9033     0.0000     2.7678     3.5481
income       0.0013     0.0541     0.0243     0.9806    -0.1049     0.1075
happy        0.3924     0.0449     8.7310     0.0000     0.3041     0.4807

************************ TOTAL EFFECT MODEL **************************
Outcome Variable: health

Model Summary:
          R       R-sq        MSE          F        df1        df2          p
     0.2163     0.0468     1.5616    24.4341     1.0000   498.0000     0.0000

Model:
              coeff         se          t          p       LLCI       ULCI
constant     4.3262     0.1574    27.4834     0.0000     4.0169     4.6354
income       0.2454     0.0496     4.9431     0.0000     0.1479     0.3429

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#overall model: Multiple R-squared (ranges from 0-1) helps us evaluate the
predictability of our model. The higher the better.

#Outcome Variable: happy => path a

#interpretation: About 26.74% of variance in happy is explained by income.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation: Income significantly and positively predicts happy. One unit
increase in income is expected to be associated with an increase of 0.6221
units in happy.

#Outcome Variable: health => path b and c'

#interpretation: About 17.35% of variance in health is explained by income
AND happy.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation b: Happy significantly and positively predicts income. One
unit increase in happy is expected to be associated with an increase of
0.3924 units in health.

#interpretation c': Income does not significantly predict health. One unit
increase in income is expected to be associated with an nonsignificant
increase of 0.0013 units in health.

#Outcome Variable: health => path c

#interpretation: About 4.68% of variance in health is explained by income
ALONE.

#individual predictor:
  #p < .05 (good);p > .05 (bad)
  #Estimate: positive (more x, more y) or negative (more x, less y)

#interpretation c: Income significantly and positively predicts health. One
unit increase in income is expected to be associated with an increase of
0.2454 units in health.

#TOTAL, DIRECT, AND INDIRECT EFFECTS OF X ON Y
#Indirect effect(s) of X on Y => a x b
#To evaluate whether M significantly mediates the relation between X and Y,
you need to check BootLLCI and BootULCI. If zero is not in the confidence
```

```
interval, then we can claim an indirect effect different from zero with 95%
confidence. This is akin to (though not exactly the same as) rejecting the
null hypothesis of no indirect effect at the alpha = 0.05 level of
significance.

#interpretation a x b: Happy significantly mediates the relation between
income and health (ab = 0.2441, 95% CI [.1823, .3121]).

#interpretation a x b: Income is significantly and indirectly related to
health through the pathway of happy (ab = 0.2441, 95% CI [.1823, .3121]).

#Conclusion: The more money you have, the happier you are, which is, in turn,
significantly associated with better health.
```

Assignment 2

Use NSCH dataset. Suppose you are interested in studying children teeth condition. Your theory is that age (**SC_AGE_YEARS**) serves as a causal antecedent variable, sugary drink consumption (**SugarDrink_21**) serves as a mediator, and the condition of children's teeth (**TeethCond_21**) serves as a consequent variable. Please run a conditional process analysis to find out the answer. (Let's assume that these three variables are measured on the same scale.). **Please submit your R script with interpretations to the designated folder on iStudy. Your R script should be titled as "assignment 2_LastFirstName."**

- Here is the information on those three variables:
  - age (**SC_AGE_YEARS**) : Higher values indicate older children.
  - sugary drink consumption (**SugarDrink_21**) : Higher values indicate that the child have sugary drinks more often on a weekly basis.
  - teeth condition (**TeethCond_21**) : Low values indicate better teeth conditions.
- Draw the diagram to represent the simple mediation model.
- Import the dataset.
- Remove value labels by using the following code (assuming you name the dataset "data"):

```
library(labelled)
val_labels(data) <- NULL
```

- Take a look at your data and descriptive statistics.
- Visualize the relation between age and sugary drink consumption . Is it positive or negative?

- Visualize the relation between sugary drink consumption and teeth condition. Is it positive or negative?
- Visualize the relation between age and teeth condition. Is it positive or negative?
- Run the conditional process analysis (model 4)
- Find out the effect of path a, b, c, c', and a x b.
- Evaluate the overall model and individual predictor.
- Write down your interpretation of the results for each path (i.e., a, b, c, c', and a x b)
- Is your theory correct? Does sugary drink consumption significantly mediate the relation between age and sugary drink consumption?

Assignment: Find an empirical journal article

- Find an empirical article in your field that uses the simple mediation model (i.e, model 4).
- What is the hypothesized model? That is, what is the diagram that visualizes the relations?
- Find out the values for a, b, c', and a x b.
- Share your interpretation of the results for each path (i.e., a, b, c', and a x b).
- Is the mediation effect significant? How do you know?
- Make a presentation to your classmates with PPT slides.
- Your presentation cannot go over than 5 mins.
- **Submit your PPT slides to the designated folder on iStudy.**